

WEATHER DATA & CLIMATE ANALYSIS

Domain Executive Briefing: Weather Domain (weather_data.csv) | Date: August 2026

1. Executive Summary

Meteorological analysis of temperature trends, rainfall distribution, and humidity correlations across various cities.

2. Key Performance Indicators (KPI Overview)

Metric Description	Key Performance Value	Strategic Benchmark / Context
Highest Avg Temp City	Bangalore	Warmest city on average
Max Recorded Rainfall	104.8 mm	Peak precipitation event
Data Quality Metric	NaNs handled safely	Missing rainfall explicitly filled with 0.0

3. Formal Statistical Hypothesis Testing

Condition Temperature Variation: --- FORMAL HYPOTHESIS TEST --- **H₀:** Mean temperature is equal across all conditions. **H_a:** Mean temperature differs by condition. Test used: One-Way ANOVA Test statistic (F-stat): 131.2676 p-value: 0.00000 95% Confidence Interval: N/A Effect Size (Eta Squared): 0.2084 Significance level: 0.05 Decision: Reject H₀ Business interpretation: Conditions have distinct temperature profiles. Post-hoc Tukey HSD reveals significant temperature differences between: Clear vs Rainy, Clear vs Sunny, Rainy vs Sunny Why this matters: Useful for forecasting energy grid demand during specific weather events. -----

4. Core Analytical & Statistical Insights

Insight 1: Finding: No significant correlation between humidity and rainfall.

Insight 2: Finding: Descriptive data indicates extreme wind speeds occur primarily during 'Storm' conditions. Meaning: Winds are conditionally associated.

5. Graphical Analytics & Visual Evidence

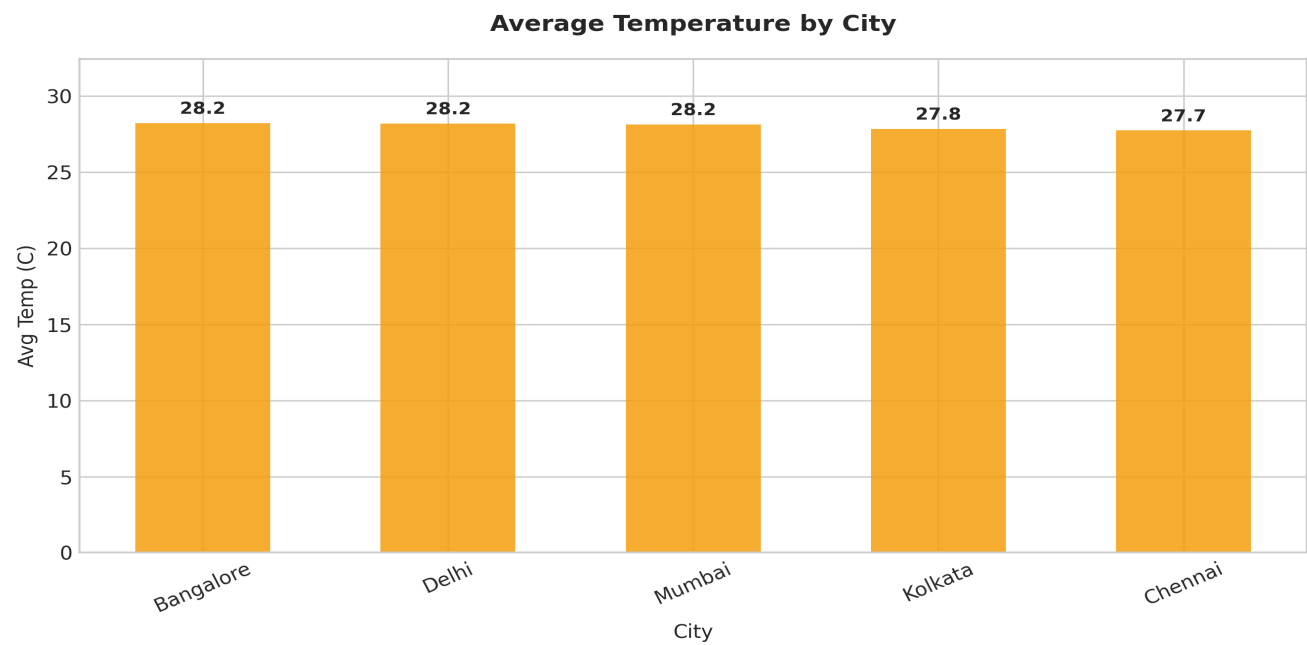


Figure 1: Visual analytical representation derived from Weather Domain (weather_data.csv) dataset.

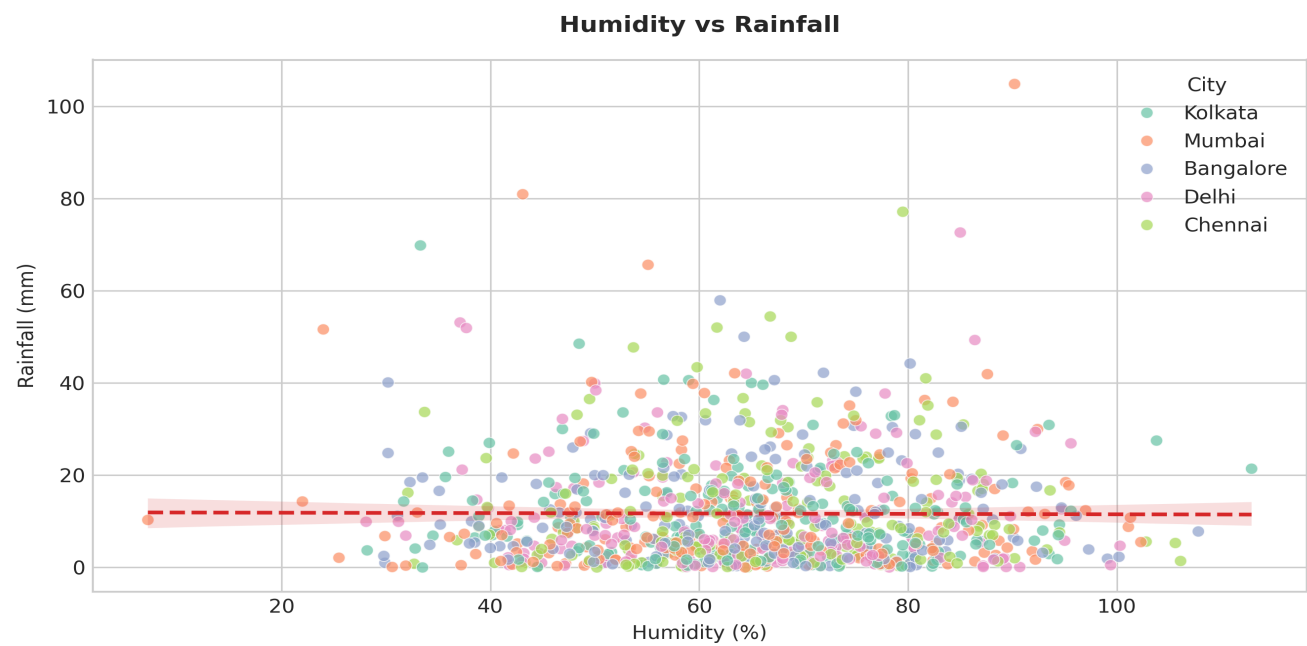


Figure 2: Visual analytical representation derived from Weather Domain (weather_data.csv) dataset.

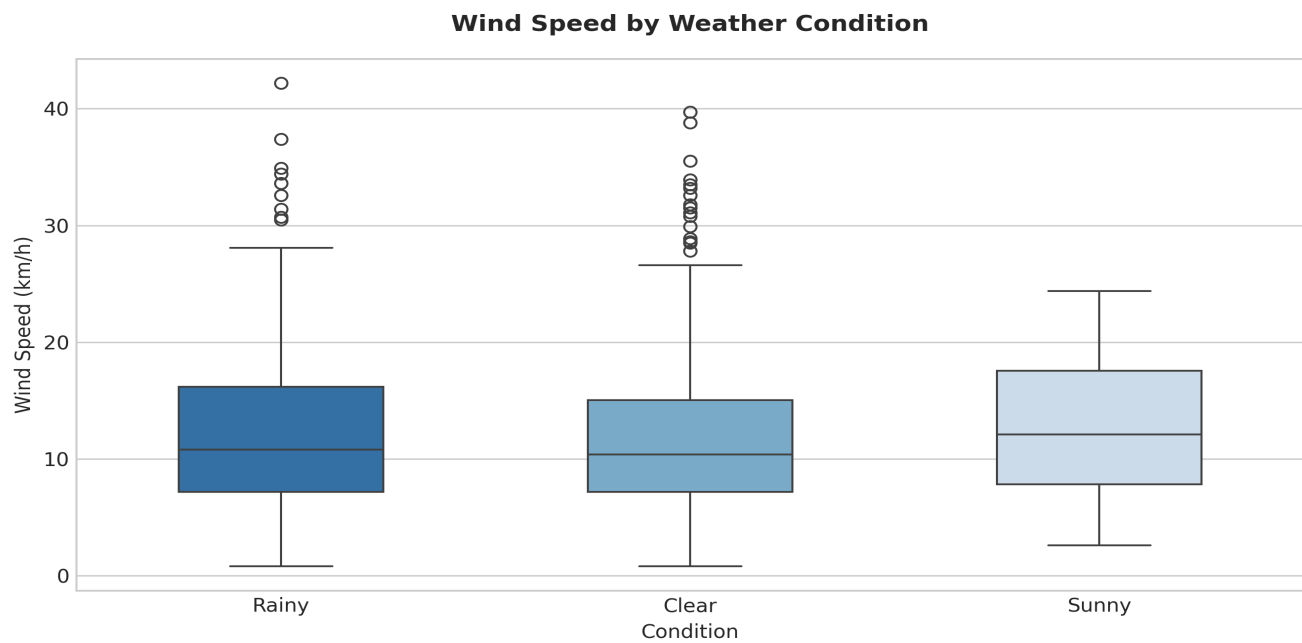


Figure 3: Visual analytical representation derived from Weather Domain (weather_data.csv) dataset.

6. Strategic Recommendations & Action Plan

Action 1: Consider evaluating localized flood warning systems when humidity exceeds 85% in coastal cities.

Action 2: Evaluate infrastructure resilience against wind gusts exceeding the 95th percentile during storm events.